

Present-Day Surface TiO_2 and Lunar Crustal Magnetic Anomalies

An underpowered test of spatial co-location, not a temporal test of an intermittent dynamo

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Independent work, openly assisted — an LLM-assisted learning project, not peer-reviewed. See §8 for the assistance and responsibility disclosure.

ABSTRACT

We compare a present-day LROC WAC surface- TiO_2 map (tio2_quantitative, ≥ 2 wt%) with the surface-evaluated Tsunakawa/Wieczorek |B| product (primary 10 nT; 25 nT sensitivity) using grouped spatial cross-validation. Binding scores use default-config XGBoost; nested tuning is diagnostic only. The full model has PR-AUC 0.395 against a matched full-model spatial-rotation null (0.2319 empirical $p = 0.138$ 86 fold-matched shifts). The H2 antipode benchmark (0.2718) is **not** recovered against its matched null (mean 0.155877, 95th 0.376621; $p = 0.172$ at $1/87 = 86+1$ resolution). Ablation drop is +0.0424 Wilcoxon p is unavailable for a significance claim. Continuous ridge $\Delta R^2 \approx -0.0146$ (report binary and continuous separately). Prevalence $\approx 3.8\%$ is the expected no-skill PR-AUC. Fitted range ≈ 403 km; $n_{\text{eff}} \approx 6.9$. Status: INCONCLUSIVE_LOW_POWER H1_Supported is false; adequate_power is false. A post-hoc USGS mare-domain sensitivity uses 30 mare blocks (15 with positives; $n = 3928$). Regenerated H1 power at strength 1.0 is 0.467 (Wilson 95% CI 0.302–0.639); adequate_power remains false.

WHAT THIS DOES AND DOES NOT SHOW

Mechanism (untouched). Nichols et al. propose a *temporal* mechanism — radiogenic melting of ilmenite-bearing cumulates at the core–mantle boundary after overturn, increasing core heat flux (not simply continued sinking). With no magnetization ages or paleointensities, this study cannot confirm or refute it; the mechanism may be entirely correct. **Mappability (tested).** A separate, narrower question — whether present-day orbital surface TiO_2 is a usable *global map proxy* for anomaly location at 30 km — for which this pipeline did not detect incremental predictive value. **Not a disproof.** The injection test recovers a strong planted TiO_2 signal only 47% of the time at the illustrative strength-1.0 anchor (target 80%) with limited effective regions, so failure to detect is not evidence of absence — hence INCONCLUSIVE_LOW_POWER, not “decoupled”. Scale/ecological inference, impact demagnetization, and downward-continuation resolution blur could each hide a real coupling in these data.

1 Scientific scope

Nichols et al. (2026) propose that deep Ti-rich cumulate melting altered core heat flow and produced short dynamo episodes associated with high-Ti volcanism. That is a claim about **timing**. A global dynamo can magnetize any rock cooling during an active interval.

This repository has no magnetization ages, eruption-age measurements, source-depth composition, or paleointensity chronology. It compares two modern maps: optical surface regolith TiO_2 and surviving crustal field strength. The historical code label “H1” therefore denotes a **candidate spatial proxy inspired by Nichols**, not the Nichols mechanism itself.

Candidate surface proxy

H2 benchmark

Present-day TiO_2 and 25/50/100 km neighborhood features. A map association would be descriptive and would not identify dynamo timing.

Distance to antipodes derived from six approximate basin centers/radii. This is a literature-motivated benchmark, not exact ground truth. It is **not** recovered against its matched H2-only rotation null ($p \approx 0.1724$; mean 0.155877; 95th 0.376621).

2 Data and measurement limits

Five 1° grids (about 30 km horizontal resolution per degree at the equator) are aligned in a lunar equal-area projection: Kaguya/Lunar Prospector surface magnetic field, LROC WAC TiO_2 , GRAIL gravity and crustal thickness, and USGS geologic units. Provenance manifests and SHA-256 hashes identify the bytes used.

The TiO_2 retrieval is a present-day optical-regolith product calibrated for mare composition. The primary footprint checks latitude, nodata, numeric range, and cross-layer coverage but is not a mare/highlands validity mask. Values in highlands are out-of-domain for this inference unless independently validated. A post-hoc USGS terrain sensitivity is reported below. The magnetic source can be older and deeper than the observed surface.

The primary target thresholds surface $|B|$ at 10 nT (25 nT sensitivity). It has about 168 positives among 4,374 quantitative Imbrian pixels (3.8%), clustered in a few provinces. Thresholding discards the continuous field magnitude; pixel count is not independent sample size.

3 Implemented analysis

Thirteen features enter logistic-regression and XGBoost baselines. The legacy repository-plan estimate uses five-fold GroupKFold over $30^\circ \times 30^\circ$ block labels, nested inner grouped folds for tuning, PR-AUC, a longitudinal-rotation null, a phase-randomized null, TiO_2 ablation, SHAP, a block bootstrap, and a block-size sweep. These are reproducible computations, but the fold construction does not establish independent regions when the fitted range exceeds the block scale. SHAP describes the fitted model; it is not evidence of a physical mechanism.

3.1 Continuous-field descriptive check

A standardized ridge comparison retains $\log_{1p}(|B|)$ rather than thresholding it. Under the same five blocked folds, controls-only R^2 is 0.2590 ± 0.1377 and controls+ TiO_2 is 0.3503. The fold-mean incremental R^2 is -0.0146 . Report binary and continuous estimands separately. This continuous result is descriptive, not independent replication: the chosen folds remain inside the fitted dependence range and no row-level p-value is reported. The small increment does not establish the surface proxy or the temporal mechanism.

4 Observed metrics

0.395 full model PR-AUC	0.2319 rotation-null mean	$p = 0.138$ descriptive null test
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Diagnostic	Value	Reading
Full XGBoost	0.395	fold SD ≈ 0.080
H2 benchmark	0.2718	matched $p = 0.172$
Full-model rotation-null mean	0.2319	$p = 0.138$
H2-only null mean / 95th	0.155877 / 0.376621	86 fold-matched rotations (pool exhausted)

Diagnostic	Value	Reading
TiO ₂ ablation drop	+0.0424	Wilcoxon <i>p</i> unavailable
Fitted range / block scale	403 / 910 km	block exceeds range; n _{eff} ≈ 6.9
Approximate effective sample size	≈1	underpowered

Both rotation nulls use fold-matched nonidentity shifts (86 accepted; pool exhausted below requested 100) and the same four evaluable folds as their observed statistics. Of 125 candidate shifts, 12 with only four evaluable folds were rejected; the first 100 eligible shifts were retained (13 eligible shifts were evaluated but unused). Accepted rotated prevalence ranges from 0.00769 to 0.02277 where the shifted target intersects the fixed Imbrian mask. The full and H2 feature sets are calibrated separately.

The H2 **non-recovery** is **fragile** in resolution: with 86 rotations the empirical-*p* resolution is $1/87 \approx 0.0099$, and the observed score clears the null 95th percentile by only 0.0037. A different rotation draw could flip nominal significance, so the result should be read as “the encoded benchmark is detectable”, not as a precise probability.

4.1 Partition sensitivity

Block	Blocks <i>n</i>	Full PR-AUC	H2 benchmark
15°	236	0.186	0.068
20°	151	0.139	0.131
30°	72	0.089	0.111
45°	32	0.049	0.085
60°	18	0.084	0.133

The sequence is non-monotone, and post-hoc K-Means folds range from 0.033 to 0.240. It would be unjustified to recast this instability as a stable negative under spatial blocking. The direct finding is that the estimate depends strongly on partition choice while the range/block diagnostic fails.

4.2 Post-hoc USGS mare-domain sensitivity

An external USGS Unified Geologic Map GIS v2 mask uses exact FIRST_Unit symbols Em, Im1, Im2, and Imd. Folds are assigned before filtering, and the comparison uses raw row-local TiO₂ only; buffers are excluded. In the Imbrian *n* quantitative scope, *n* = 3,928 across **30 mare blocks (15 contain positives)**. TiO₂+controls PR-AUC is ≈0.4766 versus controls-only ≈0.4213 (drop ≈0.0553); Wilcoxon *p* unavailable for significance.

Continuous estimands remain descriptive and are reported separately from the binary scores. Mare structure remains limited relative to CV blocks. This post-hoc result is **inconclusive**, not confirmation. The 1:5M mapped-geology mask is also a generalized proxy with source- and rasterization-dependent boundaries.

4.3 Post-hoc H2 injection-recovery power

Standalone Paper-and-Pitch power-curve JSONs are regenerated for the v2 surface product; structural adequate_power is false. Thirty phase-randomized nuisance fields per coefficient approximately preserve the spectrum of log1p-transformed observed magnetism on the real Imbrian mask and 10 nT prevalence. The encoded H2 antipode-proximity score is added at standardized latent coefficients 0, 0.5, 1, 1.5, 2, 3, and 4. The fixed XGBoost is compared with the same model after H2 ablation. Primary recovery requires a positive mean PR-AUC drop and paired one-sided *p* < 0.05 at 30°; spatially robust recovery also requires a positive drop at 60°.

Latent coefficient	Robust recovery	Wilson 95% CI
0	5/30 (0.1667)	0.0734-0.3356
0.5	11/30 (0.3667)	0.2187-0.5449

Latent coefficient	Robust recovery	Wilson 95% CI
1.0	9/30 (0.3000)	0.1666-0.4788
1.5	1/30 (0.0333)	0.0059-0.1667
2.0	0/30 (0.0000)	0.0000-0.1135
3.0, 4.0	0/30 each	0.0000-0.1135

Under the v2 surface product the H2 arm does **not** reach 80% recovery on the tested grid — robust detection peaks near coefficient 0.5 (0.37) and collapses at larger latent strengths (structurally limited; `adequate_power` remains false). Treat the grid as a design-sensitivity probe, not demonstrated realism. Fitted range ≈ 403 km and $n_{\text{eff}} \approx 6.9$; both 30° (910 km) and 60° (1,819 km) blocks exceed the fitted range.

A downward grid extension (`h2_antipode_low_strength_extension.json`; strengths 0–0.3, 30 simulations each) maps the low-strength behavior on this axis: across these strengths robust detection is only 0.100–0.367 against the 0.167 zero-strength rate, with with-control PR-AUC ≈ 0.126 –0.240. Even the H2 arm therefore has no demonstrated power at realistically-sized effects. The mean ablation drop is +0.023 **at zero strength** (removing the antipode feature hurts under pure clustered noise), which is the measured mechanism of that arm’s anti-conservative false-positive rate. The matched-null benchmark non-recovery reported above must be read in this light: a fragile rotation-test result in a score regime where the injection-based criterion fires well below target power.

The curve is a direct H2-ablation test, not power for the complete publication rule. It does not resimulate nested tuning, permutation, SHAP, or every conjunctive H1 criterion. The coefficient is a simulation unit, not a lunar-physics effect size. The artifact also leaves `target_effect_strength` null and `adequate_power` false.

4.4 Completed H1 injection curve and illustrative anchor

The same injection design applied to the **hypothesis actually under test**, with the strength grid extended down into the realistic score regime (`Paper-and-Pitch/h1_tio2_power_analysis.json`, 30 simulations per strength):

Latent strength	Robust recovery	Wilson 95% CI	With-control PR-AUC	Ablation drop
0	0/30 (0.000)	0.000–0.114	0.126	–0.0019
0.05–0.2	0–1/30 each	≤ 0.167	0.122–0.135	–0.004 to +0.004
0.4	7/30 (0.233)	0.118–0.409	0.183	+0.0354
0.6	12/30 (0.400)	0.246–0.577	0.269	+0.0881
1.0	14/30 (0.467)	0.302–0.639	0.437	+0.2142
2.0	24/30 (0.800)	0.627–0.905	0.697	+0.4261

The tested-grid 80% point estimate first appears at latent strength **2.0**, but `adequate_power` remains false because no externally justified target effect anchors the curve and the design stays structurally limited ($n_{\text{eff}} \approx 6.9$). Across the low-strength grid (0–0.2) detection sits at or below the nominal false-positive rate. Mechanistically, the ablation drop is near zero or **negative** at weak strengths because surface-TiO₂ geography is spatially collinear with the `nearside/abs_latitude` controls: after ablation, the controls absorb most of a true TiO₂-driven signal. The registered ablation criterion is therefore structurally insensitive to H1 in the weak-signal regime, a design property no seed, fold, or tuning choice repairs.

Zero-strength false-positive rates are **arm-specific**: ≈ 0.167 for the H2 arm (the smooth antipode kernel can align with clustered noise blobs spanning CV blocks) versus 0.000 (nominal) for the H1 arm under this draw. Neither arm’s operating characteristics may be quoted for the other. Under v2 the H2 arm **does not** demonstrate high-strength recovery (curve collapses above 0.5), while the H1 arm reaches point-estimate 80% only at strength 2.0 and remains at 0.467 at the illustrative strength-1.0 anchor.

Interpretive anchor (Amendment A8). No physically derived effect size exists — the theory predicts no surface-map effect — so we use latent strength 1.0 as an *illustrative anchor*. The committed artifact leaves `target_effect_strength` null. Measured H1 power at that anchor is **0.467 (Wilson 95% CI 0.302–0.639)**, far below the 0.80 requirement. `adequate_power` is therefore false **on measurement**, not by abstention, and `NOT_SUPPORTED` remains unreachable on evidence.

5 Evidence status and rule history

Status: INCONCLUSIVE_LOW_POWER — now demonstrated, not asserted. The repository-plan success criteria did not pass, so `H1_Supported` is false. The H1 injection curve measures power 0.467 (Wilson 95% CI 0.302–0.639) at the declared strength-1.0 target — a strong planted signal far above the observed score regime — and detection at realistic strengths sits at the false-positive rate. Failed H1 criteria are therefore quantitatively uninformative about H1’s truth. The observed H2 encoding is **not** recovered against its matched null.

The first adequacy-aware output used `INCONCLUSIVE_SPATIAL_AUTOCORRELATION`. After the result was observed, an asymmetric rule changed a criteria-failing run to `NOT_SUPPORTED` because leakage generally inflates scores. That post-hoc rule did not address low power. The dated amendment now reserves `NOT_SUPPORTED` for a criteria-failing run with demonstrated adequate injection-recovery power. A would-be positive that fails independence retains `INCONCLUSIVE_SPATIAL_AUTOCORRELATION`.

6 Researcher degrees of freedom

The 10/25 nT thresholds, 25/50/100 km buffers, exploratory 600→40 km gravity filter, six-basin catalogue, age masks, 30° blocks and block sweep, and model search spaces are reproducible analyst choices, not uniquely forced physical constants. The 300 km antipode length is used by the synthetic generator and H2 injection score, not the observed H2 distance feature. Optuna’s post-hoc maximum over 60 trials is uncalibrated for selection and has no inferential comparison with the repository-plan unselected null.

The configured seed now reaches XGBoost subsampling and rotation sampling. Across ten arbitrary seeds, full PR-AUC is 0.0916 ± 0.0055 and the clean Ti-derived ablation changes sign (mean $+0.0032 \pm 0.0068$). This is a software/RNG diagnostic labelled `NO_INFERENCE`, not independent lunar uncertainty.

7 Follow-up analyses

Injection-recovery power. Both arms are now complete: the direct H2 ablation curve and the H1 (TiO₂-driver) curve with a downward-extended grid. Neither is full pipeline-decision power (tuning, permutation, SHAP, and the conjunctive H1 rule were not resimulated), and the latent coefficient is a simulation unit, not a lunar effect scale. At the illustrative strength-1.0 anchor (Amendment A8; the committed artifact leaves `target_effect_strength` null), measured H1 power is 0.467 — below the 0.80 adequacy requirement, making the low-power classification a measurement.

Terrain validity. The USGS mare-domain comparison above is now available as a post-hoc descriptive sensitivity. Its small positive count and fold concentration do not provide an independent confirmation or repair the structural effective-region limitation.

8 Reproducibility is not inferential adequacy

Acquisition hashes, fail-closed ingestion, package records, deterministic seeds, and tests provide engineering rigor. They do not make pixels independent, validate highland TiO₂, or increase statistical power. The dated amendments are explicitly post hoc and preserve the original text.

This is an **independent learning project**, not peer-reviewed planetary science. Language models (LLMs) assisted with code, drafting, documentation, and adversarial review. No AI output is treated as scientific evidence; responsibility for every analysis choice and claim remains with the author. The complete decision history, including corrected mistakes, is preserved in the repository rather than hidden.

9 Conclusion

The current global rasters and pipeline do not distinguish the tested present-day surface- composition proxy from spatial null structure, and the completed H1 injection curve shows the design has measured, inadequate power (0.467 at the illustrative strength-1.0 anchor; near the false-positive rate across the low-strength grid) to detect even strong versions of that proxy. This is not a discovery of the scale at which titanium “stops” mattering, not a refutation of impact magnetization, and not a test of the temporal intermittent-dynamo mechanism. What the study now contributes is a calibrated statement of what this design **could** and **could not** have seen — the prerequisite any future co-location test must satisfy before its verdict counts.

10 References

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